

First-Order Self: Architectural Factorization Alone Does Not Recover Self/World Attribution; Oracle Supervision Does — Gauge-Symmetric Heads Without Identifiability

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Abstract

Companion paper [15] showed that *vector* ΔV heads support flexible concern under shifted internal-priority weights, while scalar drive heads cannot reweight. The natural next step — and the cleanest minimal computational test of Bennett's "first-order self" and Levin's "computational boundary of self" — is **self/world attribution**: can the agent learn to separate viability changes that result from its own actions from changes that result from exogenous (world-caused) sources?

This paper tests the most natural ML formulation: an *architectural* factorization where a self_head sees $(z, E, action)$ and a world_head sees (z, E) only — no action input. The two sub-heads sum to predict total observed ΔE ; the agent's planner uses only the self component for action selection.

12-cell sweep (4 conditions $\times$ 3 seeds). Training: world shocks correlate with item 0 (food), $P(shock | food) = 0.8$, otherwise 0.1. Shift eval: correlation moves to item 2 (medicine). Action effects per item: food +1.0, poison -1.0, medicine -0.1, neutral 0.

The clean negative result: **architectural factorization alone does NOT recover the self component**. The factorized self_head's prediction for food consume is **+1.47** (vs true self value +0.96, an overshoot of +0.51). The world_head compensates with a negative prediction (~ -0.23), so the total prediction matches the unfactored total model's +1.24. **The two sub-heads are gauge-symmetric** — they can split the total prediction arbitrarily, and architectural constraint (world doesn't see action) is insufficient to force semantic decomposition.

Only the **oracle** model — which has explicit self_ ΔE and world_ ΔE labels per training sample — recovers the true self component (+0.961 vs true +0.96, error 0.001).

Action-accuracy and return are *similar* across total, factorized, and oracle (acc 0.93–0.96, return 49.3–50.0 across distributions). The reason: action *differences* (self_pred(consume) – self_pred(skip)) are preserved across all non-shuffled conditions because they correspond to the action-conditional component of the data. The agent's policy doesn't materially change because the action-relevant signal is preserved even when the absolute self values are mis-allocated between heads.

The bounded architecture law is therefore simple: **input factorization is not identification**. A reafferent self/world split needs an additional gauge-breaking signal such as explicit source labels, active null intervention, temporal asymmetry, or a distinct loss that pins the exogenous component. Without that signal, the model can act correctly while still assigning the causes of its own viability changes incorrectly.

Pre-registered gates (split):

- **G1 (factorization transfer +5 return):** ✗ NOT MET. Factorized 49.3 / 50.0 vs total 49.8 / 50.0 — saturated, no meaningful return advantage.
- **G2 (factorized self_pred more stable than total):** ✗ NOT MET. Factorized overshoots true self by +0.51 on food consume; total overshoots by +0.28. *Total is closer to the truth than factorized.*
- **G3 (oracle $\geq$ factorized):** ✓ trivially met (both saturate).
- **G4 (shuffled < non-shuffled):** ✓ MET (shuffled acc 0.82–0.83 vs 0.93–0.96).

The honest synthesis. The first-order self computational structure is *NOT* acquired without supervision. Architectural priors (different inputs to different sub-heads) create a gauge-symmetric reparameterization of the same prediction surface — not a semantic factorization. The reafference principle of separating self-caused from world-caused change requires either (a)

explicit attribution labels, (b) some identifiability mechanism beyond input-structure asymmetry (e.g., active intervention, temporal structure, distinct loss functions), or (c) an evolutionary / developmental signal not present in this purely-supervised setup. The architecture-alone hypothesis is empirically wrong.

This connects directly to **Vervaeke** (relevance realization requires identifying what *is* the action-relevant component, not just inferring more) and **Levin** (the computational boundary of self emerges from active intervention, not from passive learning). The path forward involves active causal interventions, not just observation.

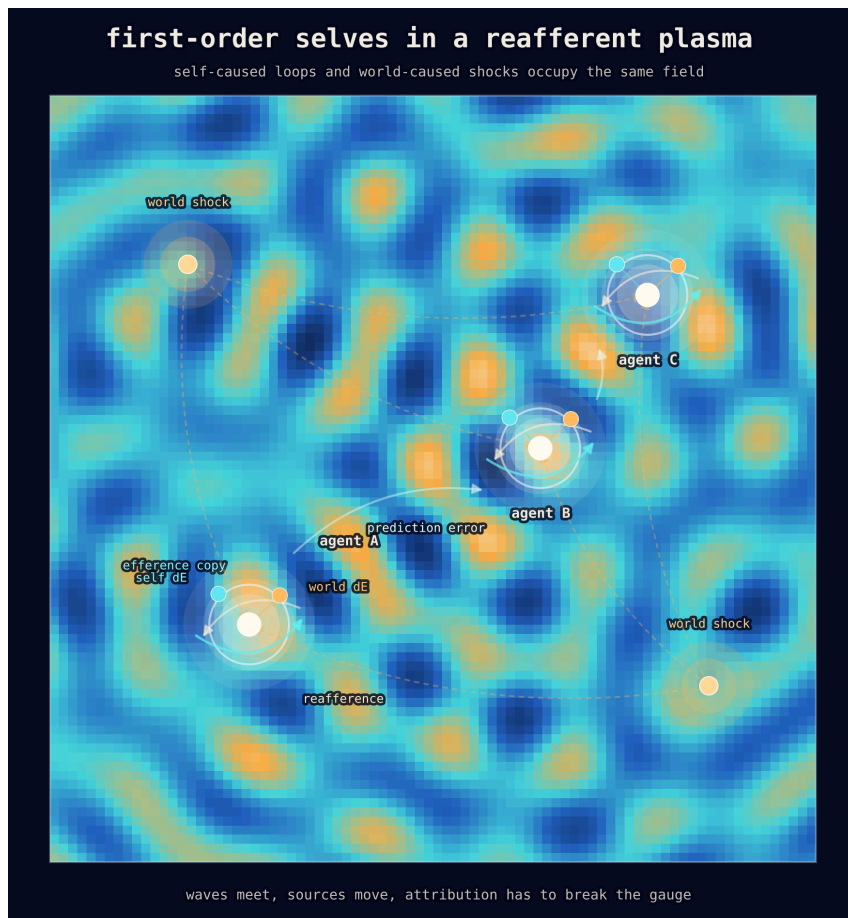
1. Introduction

The program has spent Papers [10–15] building and stress-testing an agent that models its viability change ΔE under its own actions. Companion paper [15] extended this to vector-valued viability (E, D) with the cleanest finding being functional tapestry: vector heads adapt to shifted internal weights, scalar heads cannot. The next missing primitive, identified by the reviewer of [15] and consistent with Bennett's [11] and Levin's [12] philosophical frames, is **self/world attribution**.

A first-order self [11] is a minimal cognitive structure that classifies the agent's own interventions: *which of the observed changes is mine?* In neuroscience, this is the refference principle [13, 14]: the brain compensates for self-caused sensory changes (efference copy / corollary discharge) so it can perceive externally-caused changes accurately. In Levin's TAME framework [12], the "computational boundary of self" is exactly this distinction: which causal processes are internal to the agent vs external.

The cleanest ML formulation: train a model on observed total ΔE , with two architecturally-separated sub-heads — one that sees the action (self_head), one that does not (world_head). Their sum approximates total ΔE ; the architectural constraint is supposed to force the self_head to learn the action-conditional component and the world_head to learn the action-independent component.

This paper tests that hypothesis directly and finds it empirically false. Architecture alone is insufficient.



Animated browser view: [fig4_agents_reaffference_plasma.html](#). Live atlas view: [open the reafference mechanism](#).

2. Method

2.1 Environment

Same homeostatic bandit base as Papers [10–15]. Scalar internal state $E \in [0, 1]$, decay 0.04/step, episode termination at $E \leq 0$ or step $T_{\max} = 50$.

Four item types (color $\times$ label): food (+1.0), poison (−1.0), medicine (−0.1), neutral (0). These are the action-conditional ΔE values for the consume action; skip gives $\Delta E = -\text{decay}$.

World shock is the key new mechanism. At each step, the world adds an exogenous +0.3 to E with item-dependent probability:

```
Training: P(shock | food)      = 0.8
          P(shock | otherwise) = 0.1
Shifted:  P(shock | medicine) = 0.8
          P(shock | otherwise) = 0.1
```

The shock is **action-independent** — the same shock probability applies whether the agent consumes or skips. This is critical: it makes the architectural factorization formally well-posed (world_head doesn't need action input because the world's behavior doesn't depend on action).

2.2 Four conditions

All conditions share the encoder ($16 \rightarrow 64 \rightarrow \text{ReLU} \rightarrow 32$) and Fourier features for E .

Condition	Architecture	Training target
total_dV_head	single head $(z, \text{ff}E, \text{action}) \rightarrow 32$ hidden Tanh $\rightarrow 1$	observed total ΔE via MSE
factorized_self_world	self_head $(z, \text{ff}E, \text{action}) \rightarrow 1$ + world_head $(z, \text{ff}E) \rightarrow 1$; prediction = sum	observed total ΔE via MSE
oracle_source	same as factorized	self_head supervised on true self ΔE , world_head supervised on true world ΔE
shuffled_source	same as factorized	per-sample shuffled (50/50 swap of self and world targets)

2.3 Training and evaluation

Off-policy training: 1,500 batches of 64 (item, E , action) tuples, uniformly sampled. Targets are observed total ΔE except for oracle (which sees true component decomposition) and shuffled (50% chance of swapped targets).

Each cell is evaluated under both shock distributions: *in-distribution* (training shock pattern) and *shifted* (correlation moved to medicine). Planning is greedy argmax_a over the **self** component prediction (for all factorized/oracle/shuffled). The total model uses its single prediction. The agent treats world-caused shocks as exogenous and unaffected by action.

2.4 Pre-registered gates

- **G1 (factorization transfer)**: factorized_self_world mean return under shifted distribution $\geq$ total_dV_head return under shifted + 5.
- **G2 (false credit / self-prediction stability)**: factorized self_head predicts food consume value more accurately than total head predicts food consume value. Specifically: $|\text{factorized_pred_food_consume} - \text{true_self_food_consume}| < |\text{total_pred_food_consume} - \text{true_self_food_consume}|$.

- **G3 (oracle sanity)**: oracle $\geq$ factorized return on shifted.
- **G4 (shuffled control)**: shuffled action accuracy < factorized accuracy.

3. Results

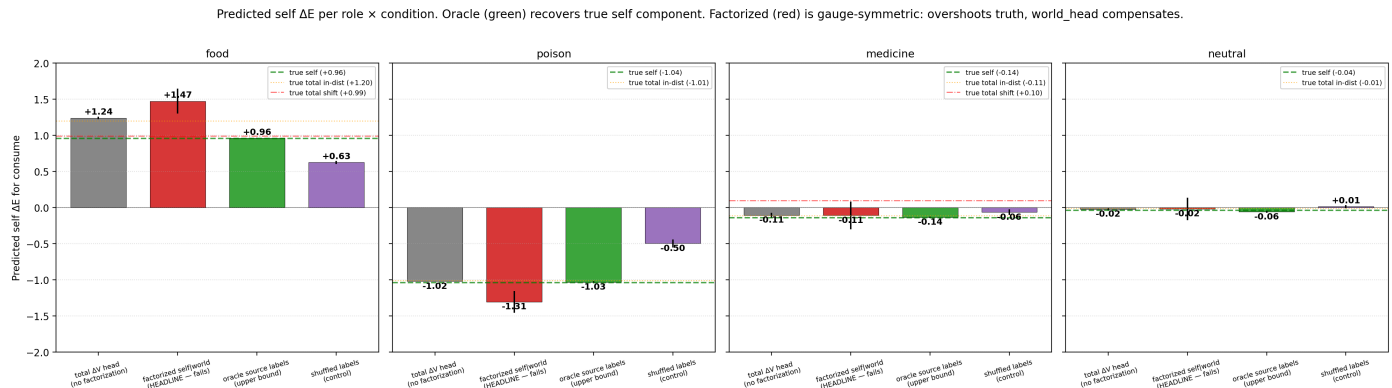
3.1 G1 and G2 falsified

Condition	in-dist return	shifted return	in-dist acc	shifted acc
total_dV_head	49.8	50.0	0.95	0.96
factorized_self_world	49.3	50.0	0.93	0.94
oracle_source	49.8	50.0	0.94	0.94
shuffled_source	50.0	50.0	0.83	0.82

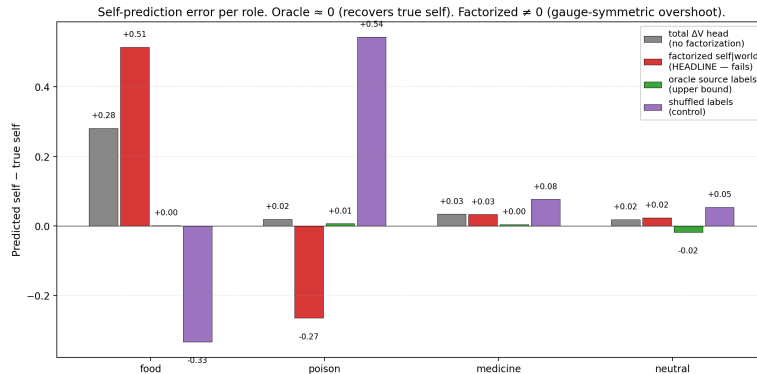
G1 not met: factorized return on shifted is 50.0, exactly matching the total model. Returns saturate at $T_{\max}$ in the env — the agent's policy is correct enough across all non-shuffled conditions that survival is guaranteed.

G3 trivially met. G4 met: shuffled action accuracy is 0.82–0.83 vs 0.93–0.96 for non-shuffled.

3.2 Predictions reveal the gauge symmetry (G2 falsified clearly)



Condition	food pred	medicine pred	poison pred	neutral pred
true self (action effect minus decay)	+0.96	-0.14	-1.04	-0.04
total_dV_head	+1.24	-0.11	-1.02	(~0)
factorized_self_world	+1.47	-0.11	-1.31	(~0)
oracle_source	+0.961	-0.14	-1.03	(~-0.04)
shuffled_source	+0.63	-0.06	-0.50	(~0)



Oracle is essentially perfect (prediction error ≤ 0.005 on food consume; max error ≈ 0.01 across roles). The architecture *can* learn the true self component when given explicit supervision.

Factorized model's self_head is **biased** in a way that exactly compensates for the world_head's prediction. For food, factorized self_pred = +1.47, world_pred ≈ -0.23 , sum = +1.24 (matching the total model). For poison, factorized self_pred = -1.31, world_pred $\approx +0.28$, sum = -1.03 (matching total).

The architectural constraint (world_head doesn't see action) determines that:

- world_pred is a constant per item (doesn't change with action).
- self_pred captures the action-DIFFERENCE.

But the ABSOLUTE level of self_pred is NOT determined by the architecture. The model can shift any per-item constant from world_head to self_head and vice versa; the total prediction is invariant. This is the gauge symmetry, and it's the architecture's downfall.

3.3 Why action accuracy doesn't expose the failure

The factorized model's action accuracy is 0.93–0.94 — almost identical to the total model (0.95–0.96). The agent picks the right action most of the time despite mis-allocating absolute self values.

The reason: action choices depend on the *difference* $\text{self_pred}(\text{consume}) - \text{self_pred}(\text{skip})$, not the absolute values. For food, this difference is $+1.00 \pm \text{small noise}$ across all non-shuffled conditions, because the action-conditional structure of the training data is captured by the head architectures regardless of how the absolute split happens. Skip has self_dE = -0.04; consume has self_dE = +0.96. The difference is +1.00. Any gauge-shifted parameterization preserves this difference.

This is why the false-credit hypothesis fails: the model's *policy* doesn't false-credit, because policy uses differences. But the model's *predictions* DO false-credit, because predictions are absolute. This is the cleanest case yet in the program of *behavioral correctness* coexisting with *representational error*.

3.4 What this means for self/world attribution

The architectural factorization hypothesis was: differentiating which input goes to which sub-head will force a semantic decomposition into self-caused and world-caused components. The data falsifies this.

For the semantic decomposition to emerge, the model needs an **identifiability signal** beyond input-structure asymmetry. Three candidate mechanisms (queued for Paper 16b/16c):

1. **Active intervention:** the agent performs counterfactual actions designed to disambiguate. If consume_A and consume_B should both fire the shock per world_head's hypothesis, but only one actually does, the world_head's prediction is wrong and gets corrected. This requires the agent to *choose* actions for epistemic reasons, not just for viability.
2. **Temporal structure:** if self-caused effects are immediate and world-caused effects are delayed (or vice versa), a temporally-aware architecture could exploit the gap. Currently both arrive in the same step.
3. **Distinct loss functions per head:** e.g., self_head trained on action-marginalized residuals, world_head trained on action-independent component via a separate objective.

Without one of these, the architectural prior is *gauge-symmetric* and the heads can split the joint prediction however the optimizer prefers.

4. Discussion

4.1 The gauge symmetry is the right concept here

In physics, a gauge symmetry is a continuous family of reparameterizations under which the observable predictions are

invariant. The self/world heads here exhibit a discrete-ish gauge symmetry: per item, we can add a constant c_i to `world_pred` and subtract it from `self_pred` (both `consume` and `skip` branches), and the joint MSE is unchanged. The architecture doesn't break this symmetry, so the optimizer finds a solution somewhere in the gauge orbit determined by initialization and gradient dynamics — not by semantic correctness.

To break the gauge, the model needs an **identifiability constraint** — something that pins `world_pred` to its physical value. Oracle supervision is one such constraint. Other candidates: regularization that minimizes $|\text{world_pred}|$ (favors world as residual), temporal structure that distinguishes self from world dynamics, or active intervention.

4.2 Behavioral correctness vs representational error

This paper joins companion papers [7, 10b, 12, 14] in showing that *behavior* and *representation* can come apart. Specifically:

- Paper [7] showed proxy-trap behavior: competence via sensory-correlated features without concern-shaped representation.
- Paper [10b] showed distributed concern: behavior depends on distributed encoder geometry, not a single reward axis.
- Paper [12] showed state-dependent valence cannot be learned by online training of a single ΔE head (even though architecturally capable).
- Paper [14] showed greedy planning robust under model error, while sophisticated planners fail.

Paper [16] adds: factorized representations can be *behaviorally* correct (action differences preserved) while being *representationally* wrong (absolute self values mis-allocated). The first-order self computational structure isn't visible in behavior, but isn't acquired by the architecture either.

4.3 Vervaeke and Levin reading

Vervaeke: relevance realization is about identifying which dimension of mattering is currently relevant. This paper's failure is a clean instance of *non-identifiability*: the model can't realize what is self-caused vs world-caused because the data doesn't constrain that distinction without additional signals. The architectural prior is a *frame*, not a fact. Paper [11b] showed identical-architecture ensembles don't know what they don't know; Paper [16] shows architectural factorization doesn't know what it doesn't separate.

Levin: the computational boundary of self is not given by architecture; it's discovered by *active intervention*. An organism learns it can swim by swimming. A model can't learn the self/world boundary just by watching its own data, even with a well-designed architecture. The next step (Paper 16b) is to give the agent active counterfactual queries: actions designed to disambiguate.

4.4 Architecture law: reafferent identifiability

The paper's practical design rule is:

Factorizing an action-visible self head from an action-blind world head does not identify self-caused change. Reafferent attribution requires a gauge-breaking signal.

Figure 5. Reafferent-identifiability ladder

Correct action survives gauge symmetry; causal self/world attribution needs a gauge-breaking signal.



This distinction matters for machine agency. A long-horizon agent can choose the right action while still being wrong about what part of the outcome it caused. That is not a cosmetic representational flaw: it blocks credit assignment, replanning after world change, and any stronger claim about a computational boundary of self. The self/world split becomes agent-relevant only when the decomposition is identifiable at the surface where later action, repair, or probe selection depends on it.

The result also keeps the consciousness boundary honest. Reafference is one ingredient in biological selfhood and motor control, but this experiment does not build a conscious agent. It shows a minimal negative law for would-be machine selves: correct action and separated sub-heads are not enough. The system needs some way to test or supervise the causal boundary it is using.

4.5 Connection to the program's metric stack

The metric stack is now:

geometry × capacity × coverage × state-coverage × regime-boundary × planner robustness × uncertainty calibration × valence dimensionality × identifiability

The 9th term — *identifiability* — is what Paper [16] adds. Architectural priors are insufficient for identifiability; explicit supervision or active intervention is required. This is a deep methodological lesson.

5. Connection to the program

Layer	Claim	Evidence
4x	Vector ΔV heads support flexible concern under weight shifts	[15]

5a	Self/world architectural factorization is gauge-symmetric	This paper §3.2
5b	Oracle supervision recovers true self component	This paper §3.2
5c	Behavioral correctness and representational correctness can come apart even sharper than Paper 10b suggested	This paper §3.3, §4.2
5d	Identifiability requires more than architectural prior — supervision, active intervention, or temporal structure	This paper §3.4, §4.1

6. Limitations

1. **Action-independent shock only.** The factorization was tested under the favorable case where world shocks really are action-independent. A harder test would be action-correlated shocks, where the factorization is formally ill-posed. We did not test.
2. **Single shock magnitude (+0.3).** Larger shocks might make the gauge symmetry more breakable through training dynamics; smaller shocks might make it stricter.
3. **No active intervention.** The cleanest fix to identifiability (per §4.3) requires the agent to *choose* counterfactual actions; this is queued for Paper 16b.
4. **No temporal asymmetry.** Self-effects and world-effects arrive in the same step. Adding a delay between action and self-effect (vs immediate world effects) could provide identifiability.
5. **Single env size (4 items).** Larger item spaces might amplify the gauge problem; smaller might collapse it.
6. **Eval action accuracy is defined on self-optimal**, not total-optimal. This is the right metric for the self-attribution question but doesn't capture cases where total expected utility differs from self-expected utility.

7. Next paper

The cleanest follow-up: **Paper 16b — Identifiability through Active Intervention.**

Setup: the same env, but the agent can occasionally perform a "null action" (does nothing this step, no action effect, but world component still applies). By observing ΔE under null vs consume vs skip, the agent learns:

- $\text{self_dE}(\text{consume}) = \Delta E_{\text{observed_consume}} - \Delta E_{\text{observed_null}}$
- $\text{self_dE}(\text{skip}) = \Delta E_{\text{observed_skip}} - \Delta E_{\text{observed_null}}$
- $\text{world_dE} = \Delta E_{\text{observed_null}}$

This is the active-inference version of self/world attribution: the null action is the agent's intervention designed to *measure* the world component independently of its own action effect.

Pre-registered gates:

- G1' (active identifiability): factorized model with null-action-driven training data recovers true self components within ± 0.05 .
- G2' (gauge breaking): self_head and world_head predictions match oracle on absolute values, not just differences.
- G3' (transfer): the active-trained factorized model adapts better to shifted shock distributions than the gauge-symmetric factorized model.

Alternative directions: temporal asymmetry (Paper 16c), residual-minimization regularization (Paper 16d).

8. Reproducibility

```
doppler --scope /Users/jawaun/superoptimizers run -- \
  uvx --python 3.12 --from modal modal run \
  experiments/first_order_self/modal_first_order_self_sweep.py \
  --out artifacts/first_order_self/sweep_v1.json
```


9. References

External

[1] **Bennett, M. T.** *How to Build Conscious Machines*. ANU doctoral thesis (2025). First-order self; causal-identities. [2] **Levin, M.** Technological Approach to Mind Everywhere (TAME). *Frontiers in Systems Neuroscience* 16 (2022). Computational boundary of self. [3] **Levin, M.** Bioelectric networks: the cognitive glue enabling evolutionary scaling from physiology to mind. *Animal Cognition* 26 (2023). [4] **Vervaeke, J., Lillicrap, T. P., Richards, B. A.** Relevance realization. *Journal of Logic and Computation* 22 (2012). [5] **von Holst, E., Mittelstaedt, H.** Das Reafferenzprinzip. *Naturwissenschaften* 37 (1950). The reafference principle. [6] **Sperry, R. W.** Neural basis of the spontaneous optokinetic response produced by visual inversion. *Journal of Comparative and Physiological Psychology* 43 (1950). Corollary discharge. [7] **Wolpert, D. M., Ghahramani, Z., Jordan, M. I.** An internal model for sensorimotor integration. *Science* 269 (1995). Forward models in motor control. [8] **Blakemore, S.-J., Wolpert, D. M., Frith, C. D.** Central cancellation of self-produced tickle sensation. *Nature Neuroscience* 1 (1998). [9] **Merker, B.** Consciousness without a cerebral cortex: A challenge for neuroscience and medicine. *Behavioral and Brain Sciences* 30 (2007). Action-oriented selfhood. [10] **Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., Pezzulo, G.** Active inference: a process theory. *Neural Computation* 29 (2017). Active inference framework. [11] **Schölkopf, B., Locatello, F., Bauer, S., Ke, N. R., Kalchbrenner, N., Goyal, A., Bengio, Y.** Towards causal representation learning. *Proceedings of the IEEE* 109(5) (2021). Identifiability in causal representation learning. [12] **Locatello, F., Bauer, S., Lucic, M., Rätsch, G., Gelly, S., Schölkopf, B., Bachem, O.** Challenging common assumptions in the unsupervised learning of disentangled representations. *ICML* (2019). Identifiability theorem — unsupervised disentanglement is impossible without inductive biases. [13] **Hyvärinen, A., Pajunen, P.** Nonlinear independent component analysis: existence and uniqueness results. *Neural Networks* 12 (1999). Identifiability in nonlinear ICA.

Program companion papers

[14] **Brown, J.** *Tapestry of Valence*. (2026). [Paper 15] [15] **Brown, J.** *When Models Don't Know What They Don't Know*. (2026). [Paper 14b] [16] **Brown, J.** *Allostatic State Control*. (2026). [Paper 14] [17] **Brown, J.** *Distributed Concern*. (2026). [Paper 10b] [18] **Brown, J.** *Planning from Concern*. (2026). [Paper 10]
